Revenue Optimisation & Business Performance Analysis in E-Commerce

Project Overview

The e-commerce industry generates enormous volumes of transactional data- purchases, pricing records, customer identifiers, and geographic origins. Yet this data remains largely underutilised for strategic decision-making, representing a significant missed opportunity for revenue growth.

THE PROBLEM

Massive transactional datasets are rarely converted into meaningful business intelligence, leaving key revenue drivers unidentified.

THE MISSION

Analyse transactional patterns to surface key revenue drivers, understand customer behaviour, and evaluate product performance across markets.

THE GOAL

Transform raw data into actionable strategic insights- enabling smarter inventory, marketing, and customer retention decisions.

Project OBJECTIVES

Trend Analysis

Examine monthly revenue patterns to identify seasonality, peak periods, and demand cycles that can inform forward planning.

Geographic Distribution

Analyse sales by country to understand market concentration and uncover international expansion opportunities.

Product Performance

Distinguish top revenue-generating products from high-volume sellers to guide assortment and pricing strategy.

Customer Insights

Evaluate individual customer contributions to revenue in order to identify high-value segments deserving focused retention efforts.

Strategic Growth

Synthesize findings into data-driven recommendations for sustainable revenue optimisation and reduced business risk.

Dataset Overview

The dataset, sourced from the Online Retail E-Commerce Dataset (Kaggle), spans a full fiscal year and provides the following analytical dimensions:

541K
Transactions
Total records across 2010-11

38
Countries
Global geographic coverage

4,223
Unique Products
Diverse SKU catalogue

4,372
Customers
Unique customer identifiers

```
# LOADING THE E-COMMERCE DATASET
df = pd.read_csv("OnlineRetail_Ecommerce_Dataset.csv", encoding="ISO-8859-1")

# DISPLAYING FIRST 5 ROWS
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.0	United Kingdom

VARIABLE	DESCRIPTION
InvoiceNo	Unique transaction identifier
StockCode	Product/item code
Quantity	Units purchased per transaction
UnitPrice	Price per unit in GBP (£)
CustomerID	Unique customer identifier
Country	Customer's country of origin
InvoiceDate	Date and time of transaction

Methodology & Analytical Approach

This project adopts a Descriptive Analytics framework, focused on interpreting historical transactional data to generate business intelligence.

Data Collection

Data Cleaning

Feature Engineering

Exploratory Analysis

Business Insights

Each stage builds sequentially upon the last- clean data enables reliable feature creation, which in turn powers meaningful EDA and, ultimately, actionable strategic conclusions.

DATA PREPARATION

Remove Cancelled Orders

All invoices prefixed with "C" were excluded, as they represent returns rather than genuine sales activity- preventing inflation of revenue figures.

Handle Missing Customer IDs

Rows with null CustomerIDs were assessed and removed where necessary to ensure accurate customer-level analysis and segmentation.

Filter Negative Quantities

Negative quantity entries- indicative of returns or data entry errors- were removed to preserve the integrity of volume and revenue.

Parse Date Columns

InvoiceDate was converted to Python datetime format, enabling month-level aggregation for trend analysis.

FEATURE ENGINEERING

The central analytical metric was engineered as follows:

REVENUE = QUANTITY × UNITPRICE

```python
# CREATING REVENUE COLUMN
df['Revenue'] = df['Quantity'] * df['UnitPrice']

df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

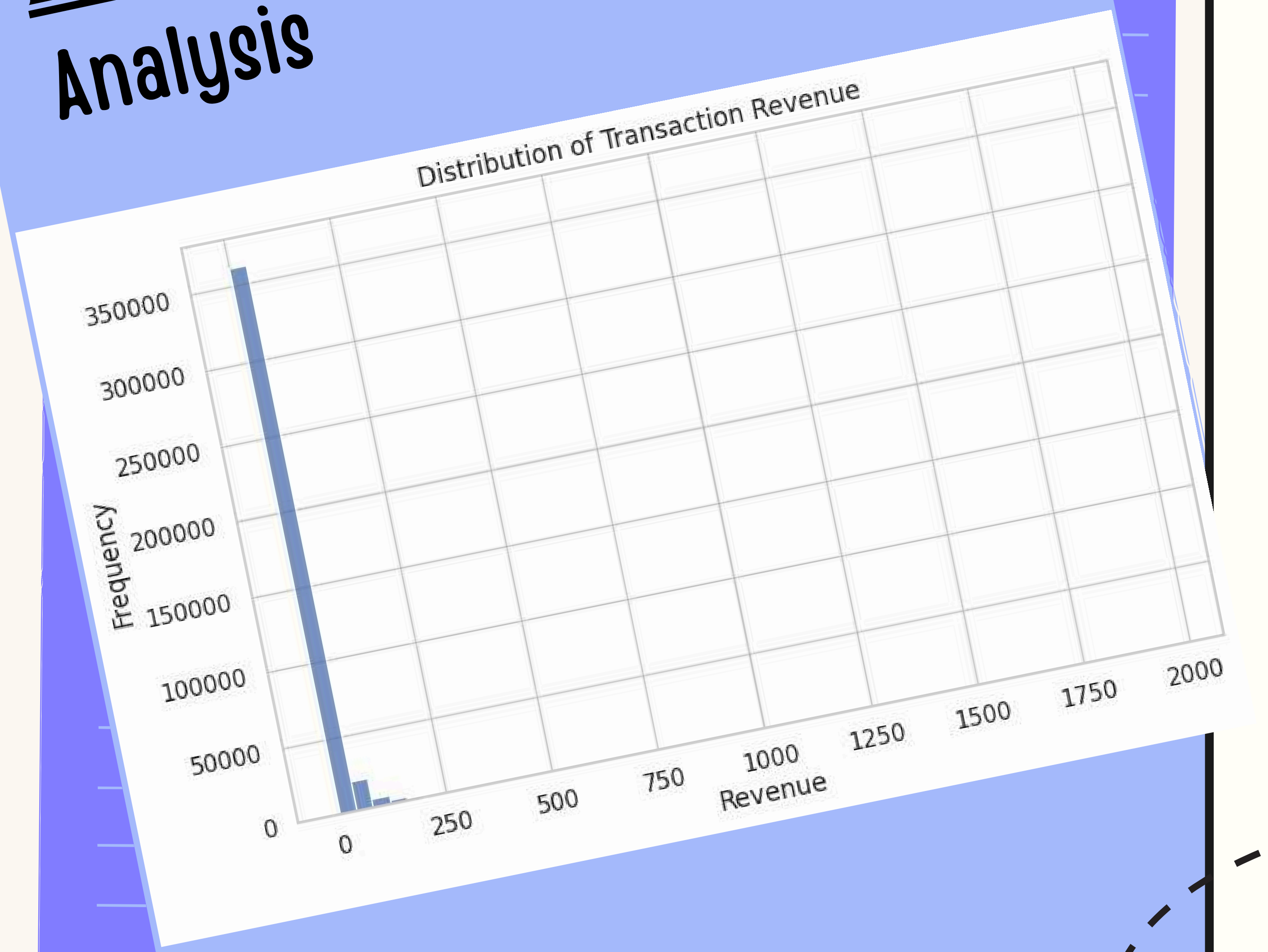
```python
# REMOVING CANCELLED ORDERS (INVOICE STARTING WITH 'C')
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]

# REMOVING NEGATIVE QUANTITIES
df = df[df['Quantity'] > 0]

# REMOVING MISSING CUSTOMER IDS
df = df.dropna(subset=['CustomerID'])

# CONVERTING DATA COLUMN TO DATETIME
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

REVENUE DISTRIBUTION
Analysis



KEY FINDING: RIGHT-SKEWED DISTRIBUTION

Analysis of transaction-level revenue shows a clear right-skewed distribution. Most transactions are below £100, while a small number of high-value orders extend the tail to the right.

This pattern is characteristic of B2B e-commerce platforms, where bulk wholesale orders co-exist with routine small purchases.

BUSINESS IMPLICATIONS

HIGH FREQUENCY, LOW VALUE

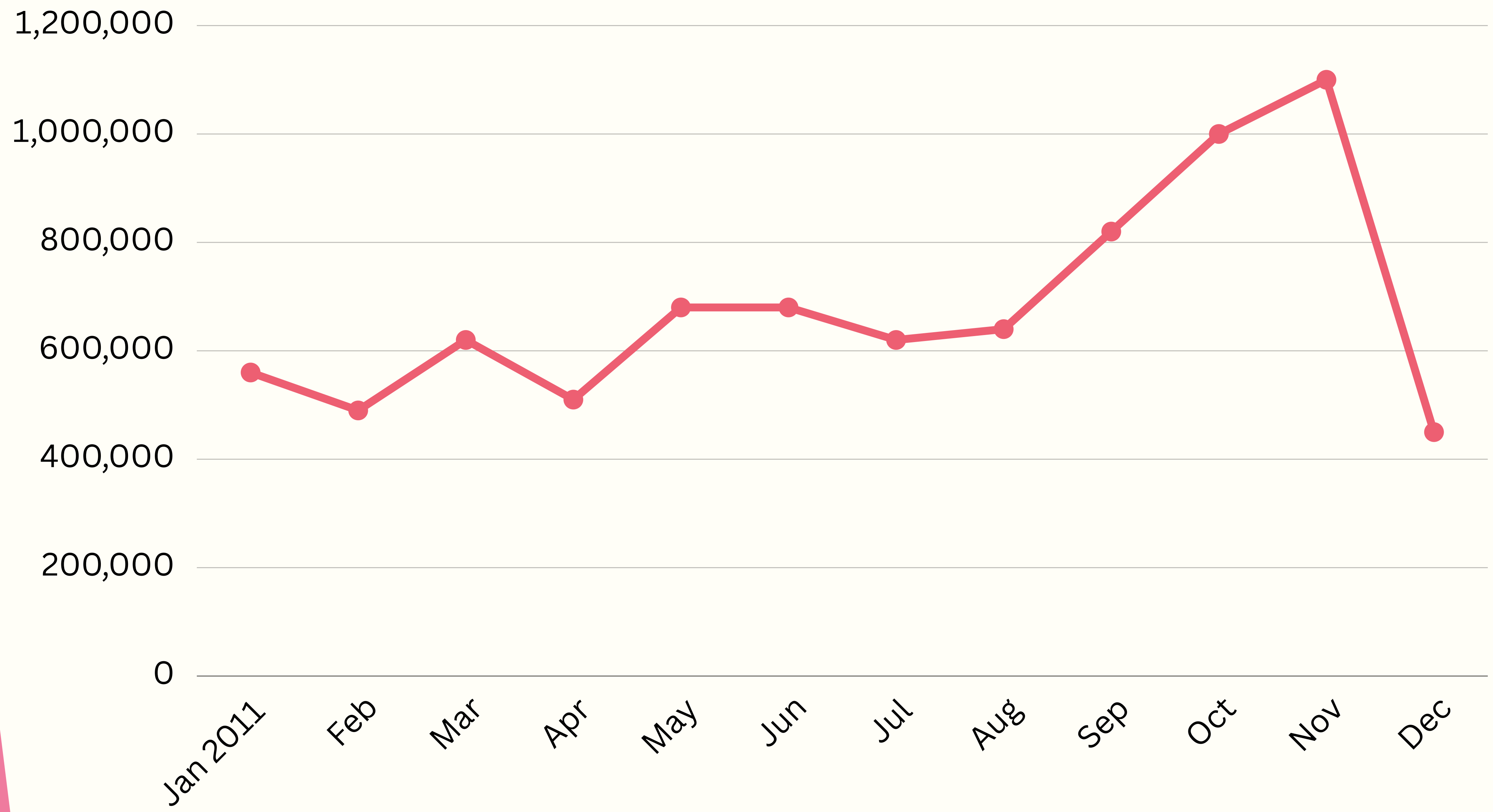
The business model relies on a very high volume of small-ticket transactions, making operational efficiency and repeat purchase rates critical drivers.

RARE BUT IMPACTFUL SPIKES

High-value orders, though infrequent, contribute disproportionately to total revenue and must be protected through dedicated account management.

MONTHLY REVENUE

Trend Analysis



Note on December: The sharp drop in December 2011 reflects incomplete data for that month- it is not indicative of a genuine performance decline. The underlying seasonal momentum remained positive.

Product Performance Analysis

A critical distinction emerges between products that generate the most revenue and those that move the most units. These two lists are not identical – and understanding the difference is essential for inventory and marketing strategy.

Strategic Insight

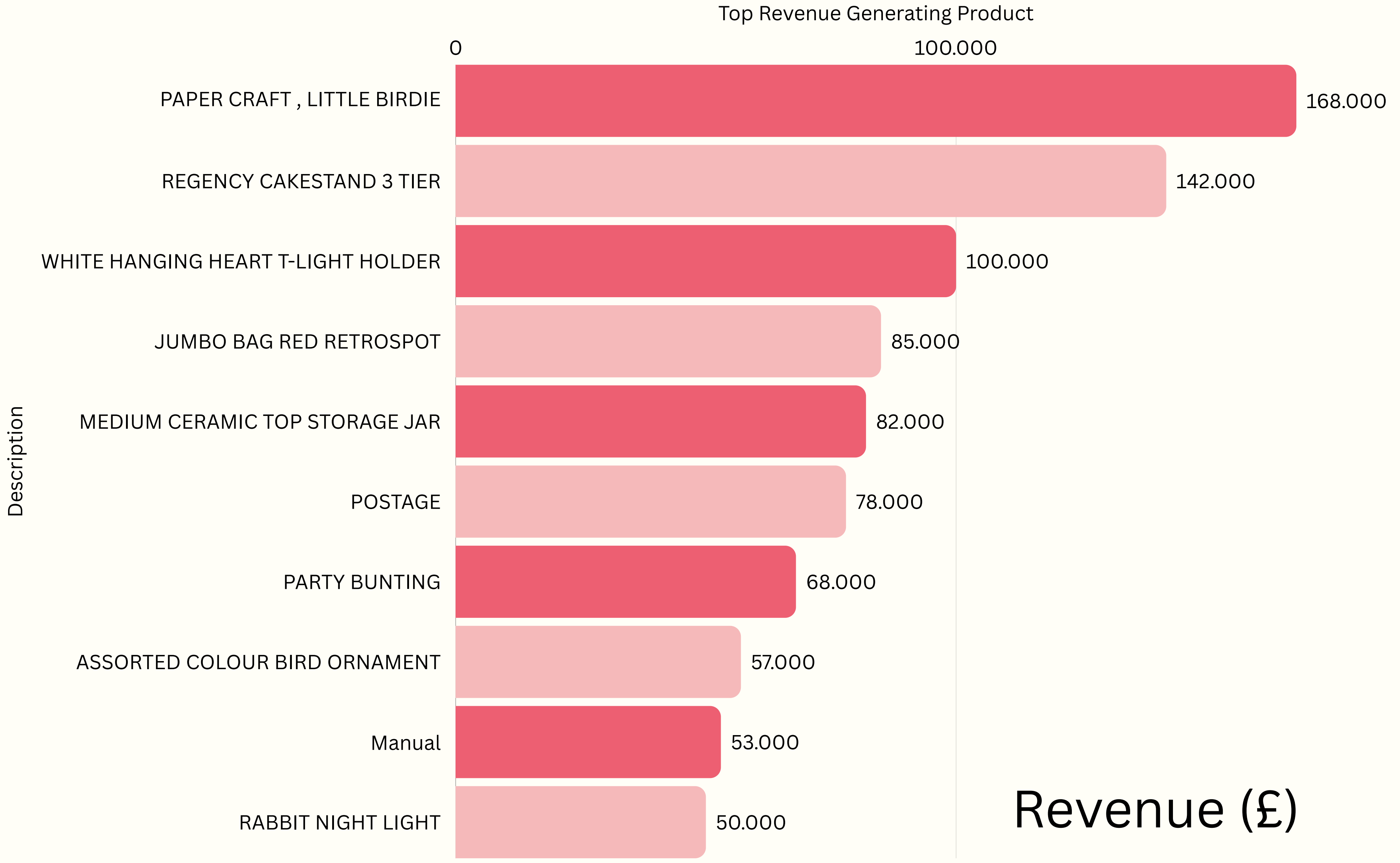
VALUE-DRIVEN PRODUCTS

High revenue per unit – prioritise for premium placement, marketing spend, and supply chain reliability.

VOLUME-DRIVEN PRODUCTS

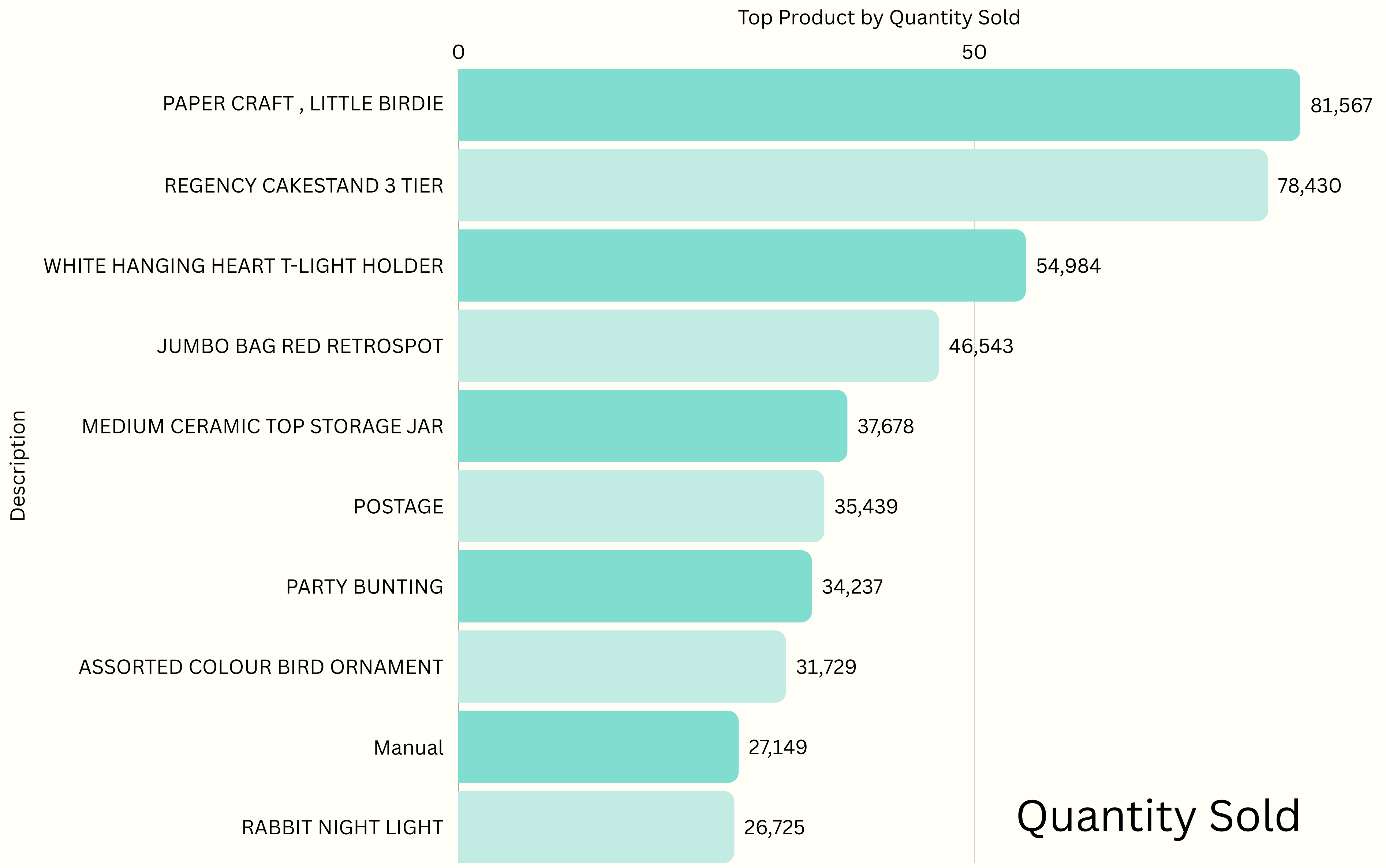
High transaction frequency – valuable for driving site traffic, basket size, and customer engagement metrics.

Champion Product: "PAPER CRAFT, LITTLE BIRDIE" generated over £160,000 in revenue – far exceeding the next-ranked item.



Product Performance & Strategic Matrix

While revenue tells us where the money is, volume tells us where the customers are. Our data reveals a critical distinction between high-margin "Value" items and high-traffic "Volume" items.



Strategic Insight

BUNDLE STRATEGY

Pair high-volume/low-cost items (Gliders) with high-value items (Cakestands) to increase Average Order Value (AOV).

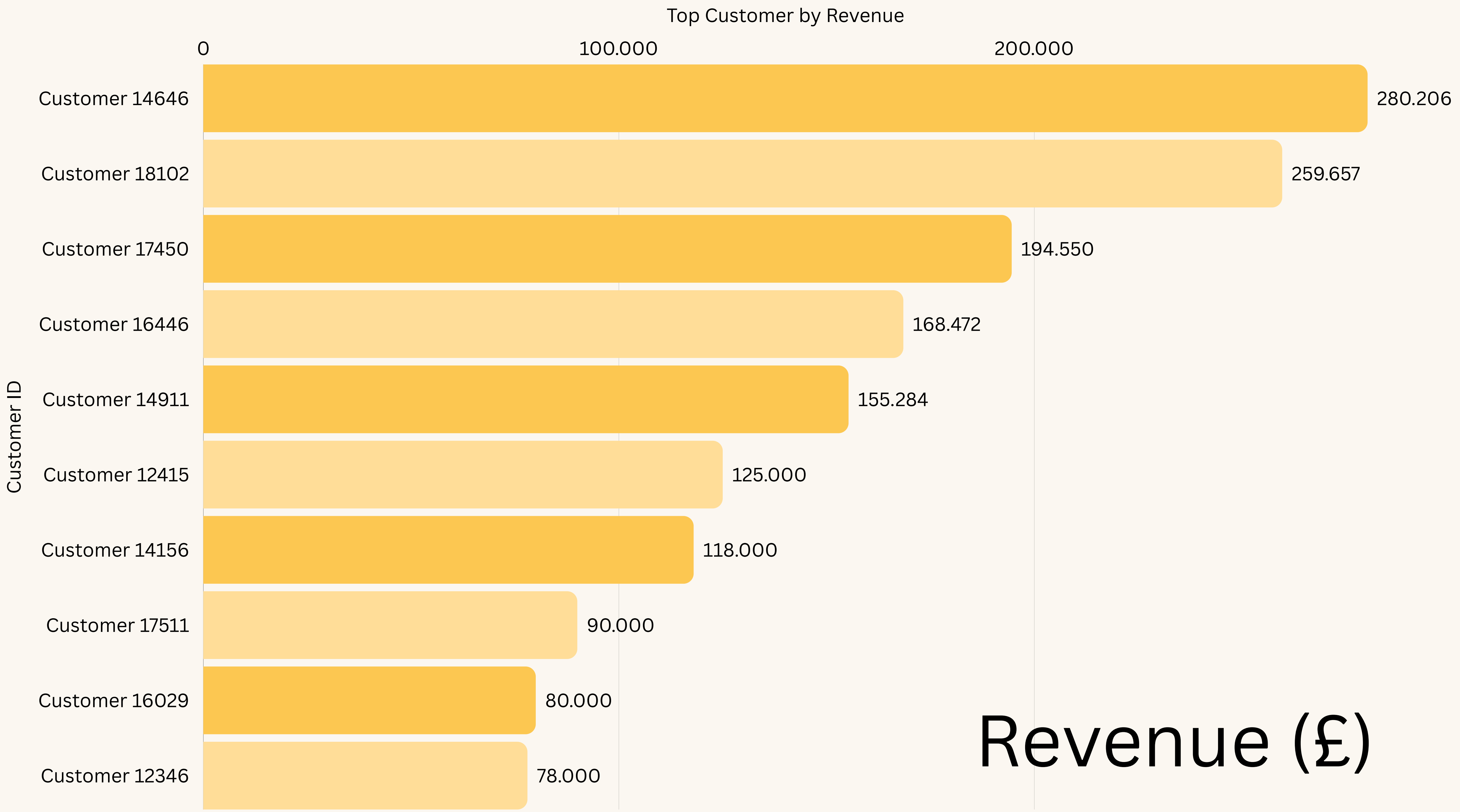
INVENTORY ALERT

Prioritize "Medium Ceramic Jars" for restocking, as they are the #2 driver for warehouse throughput.

The "Champion" (Paper Craft, Little Birdie): Dominates both revenue and volume. This is our core anchor product–protect its supply chain at all costs.

CUSTOMER CONTRIBUTION ANALYSIS

Revenue is far from uniformly distributed across the customer base. A small cohort of high-value buyers accounts for a disproportionate share of total income- a pattern with profound implications for retention strategy.



Top Customer Alone: Customer 14646 contributed approximately £280,000 highlighting extreme revenue concentration at the top of the customer pyramid.

Strategic Risk: Heavy reliance on a handful of elite customers creates vulnerability. Losing even one top account would materially impact overall revenue performance.

Average Order Value (AOV)

The Metric

£480.76

This is the Average Order Value (AOV) - the mean revenue generated per transaction after data cleaning. It reflects a relatively high per-basket spend, consistent with a wholesale or gifting-oriented customer base.

GROWTH DRIVERS TO INCREASE AOV

PRODUCT BUNDLING

Curate complementary item groupings to encourage customers to purchase multiple related SKUs in a single transaction, increasing basket size naturally.

CROSS-SELLING

Surface related or frequently co-purchased items at checkout to prompt incremental additions to the basket without friction.

UPSELLING

Present premium product variants or larger pack sizes at the point of purchase to shift customers toward higher-value options.

GEOGRAPHIC

Revenue Analysis

Geographic analysis reveals a stark imbalance in revenue distribution. Whilst international markets show engagement, the business is overwhelmingly dependent on a single domestic market.

STRATEGIC IMPLICATION

UK dependence creates risk; expanding to Germany, France, and Australia can drive growth and reduce exposure.

The United Kingdom accounts for over £7 Million in revenue – representing more than 85% of total sales. International markets contribute a fraction of this figure individually.

Key Insights & Strategic Recommendations

GEOGRAPHIC DIVERSIFICATION

Revenue is over-concentrated in the UK. Actively develop international markets – particularly Germany, Netherlands, and Australia – to reduce single-market dependency and unlock new growth vectors.

CUSTOMER LOYALTY PROGRAMMES

Top-tier customers contribute disproportionately to revenue. Implement tiered loyalty schemes, dedicated account management, and exclusive offers to retain and grow these high-value relationships.

SEASONAL INVENTORY PLANNING

Q4 demand surges are predictable and significant. Align procurement, logistics, and staffing to the November peak, ensuring supply chain readiness from September onwards.

PROMOTE HIGH-VALUE PRODUCTS

Concentrate marketing and merchandising efforts on value-driven SKUs. Apply the Pareto principle – invest in the 20% of products generating 80% of revenue for maximum ROI.

CONCLUSION & FUTURE SCOPE

CONCLUSION

This project demonstrates that descriptive data analytics can effectively bridge the gap between raw transactional logs and actionable strategic intelligence. By applying systematic data cleaning, feature engineering, and exploratory analysis to over 500,000 transactions, clear and meaningful patterns emerged – in seasonality, product performance, customer value, and geographic distribution.

FUTURE ENHANCEMENTS

Machine Learning Forecasting

Implement demand forecasting models (e.g., ARIMA, XGBoost) to predict Q4 peaks and optimise inventory pre-positioning.

Profit Margin Analysis

Integrate cost data to shift from revenue analysis to true profitability assessment – enabling margin-optimised product and market decisions.

THANK YOU!